

IMPERIAL

DEPARTMENT OF CIVIL AND ENVIRONMENTAL ENGINEERING

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Developing design formulae for flexural buckling of stainless steel columns using machine learning techniques

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Afshan and Gardner (2013) reported experimental results for cold-formed ferritic stainless steel hollow sections, building on earlier column and beam tests (Gardner and Nethercot, 2004).

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2.1.1 Column buckling and the normalised slenderness approach

Perfect columns are idealised structural members that are perfectly straight and present linear elastic, perfectly plastic behaviour. The strength of such members is bounded by two limiting modes of failure: geometric instability and material yielding. The former is governed by the Euler critical buckling load, which for a pin-ended column is

$$N_{\text{cr}} = \frac{\pi^2 EI}{L^2}, \quad (2.1)$$

where E is the Young's modulus, I is the second moment of area and L is the member length (Timoshenko and Gere, 2009). The latter is governed by the squash load $N_y = Af_y$, where A is the cross-sectional area and f_y is the yield stress. These parameters define the normalised slenderness, which collapses the column's geometry, material and length into a single dimensionless parameter, while the ultimate load N_u is normalised against the squash load to give the strength-reduction factor χ :

$$\chi = \sqrt{\frac{N_y}{N_{\text{cr}}}}, \quad \chi = \frac{N_u}{N_y}. \quad (2.2)$$

As illustrated in ??, the theoretical maximum strength of a perfect column forms a sharp χ -baseline: a pure squashing plateau ($\chi = 1$) for stocky columns and an elastic Euler hyperbola ($\chi = L/\sqrt{EI}$) for slender columns, the two regimes meeting at $L/\sqrt{EI} = 1$ (Trahair, 2008).

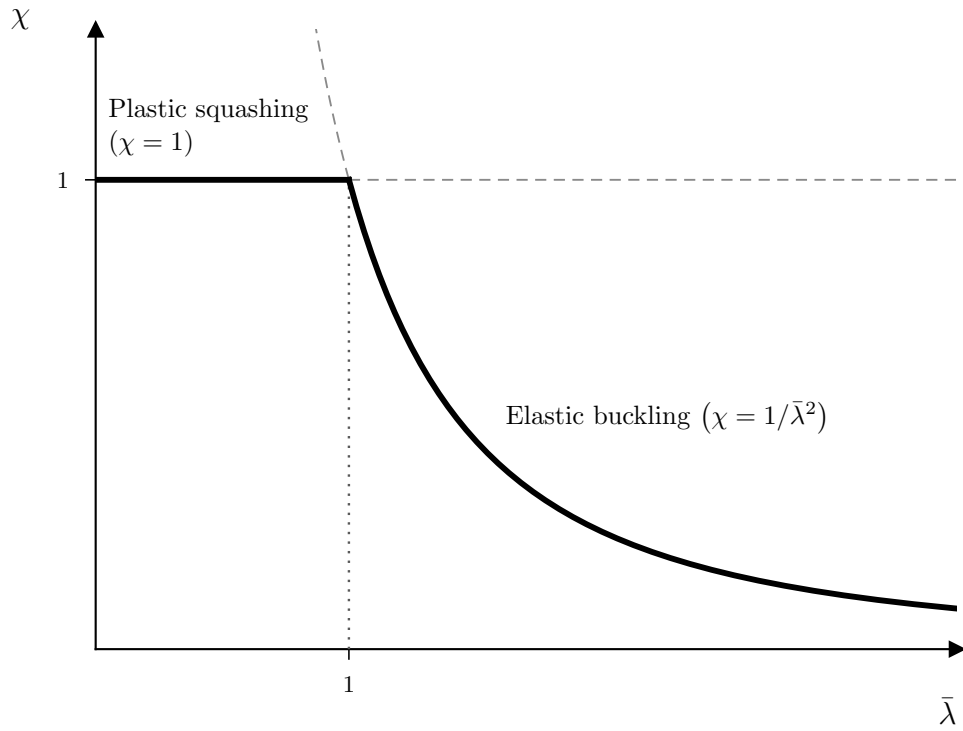


Figure 2.1: Non-dimensional buckling reduction factor χ against member slenderness for a geometrically perfect column of elastic–perfectly plastic material. The strength baseline is the lower envelope of two regimes: a plastic squashing plateau ($\chi = 1$) governing stocky members and the elastic Euler hyperbola ($\chi = 1/\bar{\lambda}^2$) governing slender members, the two intersecting at $\bar{\lambda} = 1$.

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